BENTONVILLE, ARKANSAS

KARST

Space Programming Plan

SE BENTONVILLE, AR

FEBRUARY 2026

CONFIDENTIAL

KARST — Space Programming Plan (v2)

> Detailed room-by-room square footage allocation across three scenarios. Vetted through structured founder review. This is the document Ramsey Ball requested as the foundation for site selection.
>
> "Understand the programming and the spaces you need — that drives what you're looking for in the building." — Ramsey Ball, 2026-02-26

Scenario Overview

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Scenario	Gross Building SF	Description	Construction Est. (at $500/sf)
Small (S)	~16,500 sf	Focused thermal + movement. No underground excavation — contrast therapy at grade. Compact canopy. Proof of concept / Phase 1.	~$8.3M
Medium (M)	~25,000 sf	Full vision at right scale. Underground cave. Complete programming across all levels. The target build.	~$12.5M
Large (L)	~33,000 sf	Destination scale. Generous cave, full EXOS-lite performance center, expanded food program, premium build.	~$16.5M

> **Note:** $500/sf is Ramsey's all-in construction estimate for this corridor. Numbers above are construction only — not inclusive of land acquisition, soft costs (A&E, permits ~20%), FF&E, or pre-opening reserves. See full cost summary at bottom.

Design Decisions (Confirmed)

These choices were made during the vetting process and shape every scenario:

Decision	Choice	Rationale
Sauna type	Communal Finnish sauna designed for Aufguss rituals	One large sauna, not two. Design it to support guided breathwork/towel-waving ceremonies.
Locker rooms	Fully unisex with private changing stalls	Most space-efficient and inclusive. Individual stalls, shared vanity/grooming area.
Rest areas	Integrated into widened corridors with alcoves	No dedicated relaxation room. Corridors designed with built-in benches, niches, and resting nooks between thermal areas.
Performance floor	EXOS-lite (strength + turf/agility)	No full sprint track. Focused on general population performance: racks, platforms, kettlebells, sled track, TRX, plyo.
Climbing	Bouldering feature wall (12-15 ft)	Not a roped climbing wall. Architectural drama + active bouldering with crash pads. No ropes = no staffing, lower insurance, lower ceiling requirement.
Studios	2 rooms (1 hot + 1 flex)	Hot room is dedicated. Flex room handles yoga, barre, meditation, workshops. Tight but sufficient — the Cave and performance floor are the main draws.
Food program	Cafe + CSA bodega / market	Juice bar + hot food prep + curated retail food section (local farm produce, packaged goods, supplements). Public-facing mini market. Requires commercial kitchen with hood.
Media studio	Multi-use content space	Beautiful room for podcast, video, workshops, 1:1 sessions. Not a dedicated broadcast studio. Flexible.
Event space	Library doubles as event space	No dedicated event room. Library designed with movable furniture to host talks, workshops, dinners (up to ~30-40).
Library	Signature anchor of the Canopy	Curated, immersive. 500+ volume collection. Reading nooks, chess, listening stations, living plant wall. This IS the upper level's identity.
Retail	Dedicated display area near reception	Curated KARST merch, recovery tools, supplements, skincare. 150-300 sf. Revenue stream.

Decision	Choice	Rationale
Rooftop water	None	Lounge, garden, yoga grass only. No outdoor plunge or hot soak. Simple, lower engineering risk.
Laundry	TBD — needs research	400-600+ towels/day at capacity. On-site vs. outsourced vs. hybrid to be determined. Space allocated for either option.
Staff facilities	Minimal — offices only	1-2 small offices. Staff use member facilities during off-peak. Lean operations.
Cleaning	Distributed stations per level	Locked cabinets/closets on each level with zone-specific supplies.
Restrooms (public)	Ground floor only	Near nutrition bar/reception for non-members and visitors. Members use locker rooms.
Sustainability	Full suite	Rainwater collection, greywater recycling, heat recovery from wastewater, water-efficient fixtures.
IT/Security	Dedicated server room	150-250 sf enclosed, climate-controlled. Networking, security, building management, media infrastructure.
Deliveries	Shared entrance, scheduled	No dedicated loading dock. Side door during off-peak hours.
Exterior	Cyclist-welcoming front zone	Generous bike parking, outdoor garden seating, gathering tables. Trail-facing public space designed for Greenway users.

Level 1: Underground — “The Cave”

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Function: Contrast therapy core. Sauna, cold plunge, steam, mineral baths, wet rooms. **Cell-phone-free zone. Unisex with private stalls.**

Space	S	M	L	Notes
Communal sauna (Aufguss-capable)	500 sf (30 person)	700 sf (40 person)	900 sf (50+ person)	Multi-tier cedar seating, 12 sf/person. Designed for guided rituals (towel-waving, breathwork, aromatherapy). Ref: Othership Adelaide 640 sf / 50 people
Cold plunge stations	300 sf (4 stations)	500 sf (6 stations)	700 sf (8-10 stations)	Pools + wet deck + circulation. Two temps: 45°F and 55°F. ~96-112 sf water surface per large pool + 2.5-3x for deck/access
Steam room	150 sf (10 person)	250 sf (15-20 person)	350 sf (25 person)	Eucalyptus-infused. Stone-tile walls. 8-12 sf/person at 8ft ceiling
Mineral soak pool(s)	—	350 sf (1 pool, 10-12 person)	500 sf (2 pools, 15-20 person)	Warm mineral water, 95-102°F. Natural stone construction. Cut in S scenario
Wet rooms / showers	400 sf (12 stations)	600 sf (16-20 stations)	800 sf (24+ stations)	All private stalls. Mix of rain and handheld. Stone/tile with drainage
Changing / lockers	500 sf	800 sf	1,100 sf	Fully unisex. Individual private changing stalls. Shared grooming/vanity area. Full-size lockers (80-120+)
Transition corridors with rest alcoves	350 sf	600 sf	850 sf	Widened stone corridors with built-in benches, niches, resting nooks. Riverstone floors, ankle-level path lighting. Rest integrated into the journey, not a separate room
Cleaning station	25 sf	40 sf	50 sf	Locked closet with wet-zone cleaning supplies, slop sink
Cave subtotal (net usable)	**2,225 sf**	**3,840 sf**	**5,250 sf**	

S scenario note: No underground excavation. The “cave” programming is at grade in an enclosed, windowless section with cave-like design treatment (stone, dim lighting, cedar). Dramatically reduces construction cost and geotechnical risk.

Comparable benchmarks:

- Othership Adelaide (Toronto): 3,000 sf — 50-person sauna, 4 plunges, tea room
- Sauna House (Charlotte): ~5,000 sf — full thermal suite
- Othership Yorkville (Toronto): 5,000+ sf — 90-person sauna, 8 plunges

Level 2: Ground — “The Foundation”

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Function: Movement, training, nutrition, community entry point. Public-facing elements.

Performance Training Floor (EXOS-Lite)

Space	S	M	L	Notes
Strength floor	2,500 sf	3,500 sf	5,000 sf	Racks, platforms, free weights, kettlebells, machines. Premium equipment (Eleiko, Rogue) integrated into architecture. Ref: EXOS allocates 7-10K sf at flagship scale
Functional / turf / agility area	1,000 sf	2,000 sf	3,000 sf	Open floor: sled track, plyo boxes, TRX rig, medicine balls, movement flow. Sprung floor. No full sprint lanes — focused athletic training for general population
Performance subtotal	**3,500 sf**	**5,500 sf**	**8,000 sf**	

Studios

Space	S	M	L	Notes
Studio 1 — Hot yoga / breathwork	600 sf (20 person)	700 sf (25 person)	800 sf (28 person)	Sealed, heated 95-105°F, infrared + traditional heat, very dim. 25-30 sf/person. Womb-like. Ref: Yoga Tropics aesthetic
Studio 2 — Flex (yoga/barre/meditation/workshops)	600 sf (20 person)	700 sf (25 person)	900 sf (30 person)	Multi-use: yoga, barre, movement, meditation, workshops. Wall-mounted barre. Mirrors optional. Lighting programmable. 21-25 sf/person
Studios subtotal	**1,200 sf**	**1,400 sf**	**1,700 sf**	

Bouldering Feature

Space	S	M	L	Notes
Bouldering wall + crash pad zone	500 sf	650 sf	800 sf	12-15 ft textured wall. Natural stone aesthetic, not plastic-gym look. Crash pads below. Self-serve, no staffing needed. Dramatic visual feature visible from entrance. No ropes = lower ceiling requirement (18-20 ft clear), lower insurance
Bouldering subtotal	**500 sf**	**650 sf**	**800 sf**	

Cafe, Market & Nutrition

Space	S	M	L	Notes
Kitchen / prep (back-of-house)	350 sf	500 sf	650 sf	Commercial kitchen with hood for hot food. Juicers, blenders, prep stations, fridges, dishwashing. 40% of total food program footprint
Counter / service / POS	150 sf	200 sf	250 sf	Counter service, display cases, grab coolers, queue space
CSA bodega / retail food market	200 sf	300 sf	400 sf	Curated local farm produce, packaged goods, supplements, health foods. Public-facing, no membership required
Cafe seating (indoor)	100 sf (6 seats)	150 sf (8-10 seats)	200 sf (12 seats)	Counter-height and table seating. Integrated with reception area flow
Cafe/market subtotal	**800 sf**	**1,150 sf**	**1,500 sf**	

Reception, Retail & Public Areas

Space	S	M	L	Notes
Reception / community gathering	400 sf	600 sf	800 sf	Welcome desk, community seating, daily programming display. Lodge/home feel. Warm materials — timber, stone
Retail display area	150 sf	200 sf	300 sf	Branded KARST merch, recovery tools (Therabody, Hyperice), supplements, skincare. Curated, not cluttered
Public restrooms	150 sf	200 sf	250 sf	Near reception/cafe for non-members. ADA compliant. 2-3 stalls + sinks
Management offices	150 sf (1 office)	250 sf (2 offices)	300 sf (2 offices)	Small offices for admin, scheduling, meetings. Staff break integrated with cafe BOH
General storage / back-of-house	150 sf	250 sf	400 sf	Equipment storage, retail stock, seasonal items
Laundry / linen staging	100 sf	200 sf	300 sf	If on-site laundry: commercial washer/dryer, folding, linen storage. If outsourced: staging closet for daily pickup/delivery. Space allocated for either. TBD
Cleaning station	25 sf	40 sf	50 sf	Locked closet, ground-level supplies
Reception/ops subtotal	**1,125 sf**	**1,740 sf**	**2,400 sf**	

Ground Level Total

Category	S	M	L
Performance floor	3,500 sf	5,500 sf	8,000 sf
Studios	1,200 sf	1,400 sf	1,700 sf
Bouldering	500 sf	650 sf	800 sf
Cafe/market	800 sf	1,150 sf	1,500 sf
Reception/ops	1,125 sf	1,740 sf	2,400 sf
Ground total	**7,125 sf**	**10,440 sf**	**14,400 sf**

Level 3: Upper — “The Canopy”

Function: Mind, media, learning, community. The intellectual and creative layer.
The library is the anchor.

Space	S	M	L	Notes
Library / sound lounge (doubles as event space)	900 sf (25 person)	1,100 sf (30 person)	1,300 sf (35 person)	Signature space. 500+ volume curated collection (health, philosophy, science, design). Reading nooks, chess tables, listening stations, living plant wall. Movable furniture allows reconfiguration for talks, workshops, community dinners (up to ~30-40). A/V system for presentations. 30-35 sf/person lounge layout
Multi-use content / media room	300 sf	400 sf	500 sf	Beautiful room that works for podcast recording, video production, small workshops, 1:1 sessions. Not a dedicated broadcast studio — flexible. Sound treatment, good lighting, camera-friendly. Visible through glass wall from library
Practitioner / PT rooms	200 sf (1 room)	400 sf (2 rooms)	400 sf (2 rooms)	200 sf each: treatment table, sink, storage. Sound-isolated. Revenue share with practitioners
IT / server room	100 sf	175 sf	250 sf	Climate-controlled. Networking, security DVR, building management system, media infrastructure. Fiber uplink
Cleaning station	25 sf	40 sf	50 sf	Locked closet, upper-level supplies
Circulation / stairs / elevator	250 sf	350 sf	450 sf	Feature staircase connecting all levels. Elevator for ADA access
Canopy subtotal	**1,775 sf**	**2,465 sf**	**2,950 sf**	

Level 4: Rooftop — “The Sky”

Function: Outdoor recovery, views, garden, movement. Journey culmination. No water features.

Space	S	M	L	Notes
Outdoor lounge	400 sf	700 sf	1,000 sf	Recovery seating, daybeds, shade structures (timber pergola + retractable fabric), fire feature for cool weather. Ozark hill views
Grass strip (outdoor yoga/movement)	300 sf (10 person)	500 sf (15 person)	800 sf (20-25 person)	Real or premium synthetic grass. Barefoot movement. Morning yoga, evening stretching
Rooftop garden	200 sf	400 sf	600 sf	Raised timber beds. Herbs/edibles feeding cafe below. Stormwater retention. Corten steel or raw timber planters
Rooftop subtotal	**900 sf**	**1,600 sf**	**2,400 sf**	

Mechanical & Infrastructure

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This is the space most commonly underestimated. Thermal facilities require massive mechanical infrastructure — water heating, filtration, humidity control, chemical treatment, pumps — far beyond standard commercial HVAC.

Water Systems Mechanical

Space	S	M	L	Notes
Central water plant	600 sf	1,000 sf	1,400 sf	Boilers, heat exchangers, filtration (sand/glass media), UV/ozone sanitation, recirculation pumps, surge tanks. Industry rule: 50 sf per 10,000 gallons served, OR 25% of total water surface area — whichever is larger. 6 ft clearance around major equipment for maintenance. 8+ ft ceiling for overhead pipe manifolds
Chemical storage	75 sf	100 sf	150 sf	Dedicated room. Ventilated directly to exterior. Oxidizers and acids NEVER co-located (produces chlorine gas). Secondary containment basins. Below 95°F. Per OSHA HazCom + NFPA 30
Heat recovery / sustainability	—	100 sf	200 sf	Wastewater heat recovery (HUBER RoWin or equivalent). Reduces heating energy 35-50%. Rainwater collection controls. Greywater treatment. Needs gravity feed — argument for sub-grade placement

Placement recommendation: Central water plant should be adjacent to (or below) the Cave for shortest pipe runs, gravity drainage, and noise isolation. In the S scenario (cave at grade), the mechanical room is adjacent at grade. In M and L scenarios with underground cave, mechanical is also underground. This is the dominant pattern at Bathhouse Brooklyn, Othership, and Therme Group facilities.

Critical design note: Each thermal zone (pool hall, steam, sauna, transition) needs a dedicated air handling unit with separate exhaust path. Shared HVAC systems between wet and dry zones cause humidity migration and are the most common commissioning failure in thermal facilities. Pool-grade dehumidification (PoolPak PPK or Dectron) is mandatory — standard commercial HVAC will fail.

Building Systems

Space	S	M	L	Notes
Building HVAC / electrical	300 sf	500 sf	700 sf	Standard building mechanical (non-water). Electrical panels, HVAC for dry zones, fire suppression
Infrastructure subtotal	**975 sf**	**1,700 sf**	**2,450 sf**	

Exterior / Streetscape

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Not counted in building SF, but critical for site selection and brand experience.

Space	S	M	L	Notes
Cyclist welcome zone	400 sf	600 sf	800 sf	Generous covered bike parking (20-40+ bikes), repair station, direct Greenway trail connection. Benches, water fountain. The FIRST thing trail users see
Garden courtyard with gathering tables	400 sf	800 sf	1,200 sf	Native Ozark plantings. Communal tables and seating for outdoor gathering. Water feature optional. Connects building to Greenway and neighborhood. Designed for lingering, not just passing through
Outdoor cafe seating	200 sf	300 sf	500 sf	Street-facing, extends nutrition bar / bodega outdoors. Casual, public, welcoming
Art court (basketball/pickleball)	—	2,000 sf	3,000 sf	Custom art surface design. Public-facing, lit for evening. Community activation. Cut in S
Stormwater / landscape	Integrated	Integrated	Integrated	Permeable paving, rain gardens, bioswales. Part of sustainability suite
Exterior subtotal	**~1,000 sf**	**~3,700 sf**	**~5,500 sf**	

Building Totals Summary

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Net Usable Area

Level	S	M	L
Underground — The Cave	2,225 sf	3,840 sf	5,250 sf
Ground — The Foundation	7,125 sf	10,440 sf	14,400 sf
Upper — The Canopy	1,775 sf	2,465 sf	2,950 sf
Rooftop — The Sky	900 sf	1,600 sf	2,400 sf
Mechanical & Infrastructure	975 sf	1,700 sf	2,450 sf
Net usable subtotal	**13,000 sf**	**20,045 sf**	**27,450 sf**

Gross Building Area (with circulation, walls, structure @ ~25%)

	S	M	L
Net usable	13,000 sf	20,045 sf	27,450 sf
Circulation / walls / structure (25%)	3,250 sf	5,011 sf	6,863 sf
Gross building area	**~16,250 sf**	**~25,056 sf**	**~34,313 sf**

Exterior (not in building SF)

	S	M	L
Exterior program	~1,000 sf	~3,700 sf	~5,500 sf

Cost Estimate Summary

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All based on Ramsey's $500/sf all-in construction estimate for this corridor.

	S	M	L
Construction (building @ $500/sf)	~$8.1M	~$12.5M	~$17.2M
FF&E (equipment, sauna systems, commercial kitchen, furniture)	~$700K	~$1.2M	~$1.8M
Soft costs (A&E, permits, engineering @ 20% of construction)	~$1.6M	~$2.5M	~$3.4M
Pre-opening (operating reserve + marketing)	~$1.0M	~$1.3M	~$1.5M
Total project cost (excluding land)	**~$11.4M**	**~$17.5M**	**~$23.9M**

> **Land is additional.** In this corridor, land runs $40-81/sf per Ramsey's recent deals. A 1-acre site (~43,560 sf of land) at $60/sf midpoint = ~$2.6M for land alone. 407-409 F St is asking ~$3.5M.

> The Medium scenario at ~$17.5M + ~$3M land = ~$20.5M total. This aligns with (and slightly exceeds) the earlier $16-21M range, which Ramsey called "expensive but not necessarily wrong."

What Gets Cut in Each Scenario

Small (S) — Proof of Concept (~16,250 sf)

- No underground excavation (thermal suite at grade in enclosed section)
- No mineral soak pools
- No art court
- Smaller of everything — 2 studios, compact cafe, 1 practitioner room, 1 office
- Could work as the “pop-up test” Ramsey recommended, if built as a lighter intervention in an existing space

Medium (M) — The Target Build (~25,000 sf)

- Underground cave with full thermal suite including mineral soak
- EXOS-lite performance center with turf/agility area
- Full cafe + CSA bodega
- Signature library as Canopy anchor
- Art court + full exterior program
- This is the vision at the right scale

Large (L) — Destination Scale (~34,000 sf)

- Everything in Medium, plus:
- Expanded performance floor (8,000 sf — serious EXOS-class territory)
- Larger thermal suite with dual mineral pools
- Bigger cafe/market operation
- Expanded content/media room
- Full outdoor campus with larger art court
- Room to grow into over 10+ years

Capacity at Each Scenario

Metric	S	M	L
Max simultaneous occupancy	~100	~180	~270
Daily throughput estimate	~350-450	~600-800	~900-1,100
Membership capacity (7-10x simultaneous)	700-1,000	1,250-1,800	1,900-2,700
Contrast therapy sessions/day	~80-120	~200-300	~350-450

Site Selection Criteria (Derived from Programming)

Requirement	S	M	L
Building footprint (if single story)	~8,500 sf	~13,000 sf	~17,500 sf
Multi-story footprint	~5,500 sf (3 floors)	~7,000 sf (3 floors + basement)	~9,000 sf (4 floors + basement)
Ceiling height (main performance area)	18-20 ft min (bouldering)	18-20 ft min	20-25 ft preferred
Below-grade capability	Not required	Required (cave)	Required (cave + mechanicals)
Outdoor space needed	~1,200 sf	~4,000 sf	~6,000 sf
Parking	25-35 spaces	45-65 spaces	70-90 spaces
Trail adjacency	Preferred	Preferred	Strongly preferred
Lot size (building + outdoor + parking)	~0.5 acre	~0.9 acre	~1.3 acre
Water/sewer capacity	High (thermal)	Very high	Very high
Electrical capacity	High (sauna, kitchen, HVAC)	Very high	Very high

Priority Parameters (Per Ramsey's Advice)

1. **Trail-side?** Preferred but not mandatory — adds value for brand and Greenway cyclist traffic
2. **SE Market District?** Strong preference — Ice House synergy, 8th Street Market, The Momentary ecosystem

3. **Open to adjacent areas?** Yes — further west, further south (102 area), Bella Vista (concept evolves)
4. **Embedded in larger development?** Yes — Ramsey's recommendation. Position KARST as amenity within a mixed-use project
5. **Freestanding vs. within a building?** Either — freestanding is expensive in this corridor but gives maximum design control

Open Items for Follow-Up

Item	Status	Notes
Laundry strategy (on-site vs. outsourced)	TBD	Estimated 400-600+ towels/day at capacity. Space allocated for either. Research needed on cost comparison
Mechanical room placement (split vs. centralized)	Recommendation: centralized, sub-grade	Per industry best practice for thermal facilities. Final decision depends on site geotechnical conditions
Water treatment specialist	Needed	Hire aquatics/spa mechanical engineer (e.g., Counsilman-Hunsaker) before finalizing mechanical sizing
GWI Hydrothermal Guide	Download	Global Wellness Institute's Guide to Hydrothermal Spa Development Standards (4th Ed, 2024). Free. Industry reference document for this building type
ADA compliance audit	Needed	All levels accessible by elevator. Wet-zone ADA requirements for thermal facilities are specific and strict
Fire/egress code review	Needed	Underground occupancy has specific egress requirements. Two means of egress from Cave likely required

Comparable Facility Sizes (Reality Check)

Facility	Size	Pricing	Market
Othership Adelaide, Toronto	3,000 sf	$244 CAD/mo	Dense urban
Sauna House Charlotte	~5,000 sf	$45-59/session	Secondary market
Othership Yorkville, Toronto	5,000+ sf	$244 CAD/mo	Dense urban
SALT Health, Bentonville	12,300 sf	$349/mo	NWA
BlakeSt., Bentonville	21,380 sf	$155-255/mo	NWA
Bathhouse Brooklyn	28,000 sf (post-expansion)	$50-65/day, $145-225/mo	NYC
EXOS Phoenix HQ	31,000 sf	Corporate/pro	Phoenix

Our Medium scenario (~25,000 sf) sits between BlakeSt. (21K) and Bathhouse Brooklyn (28K) — the right neighborhood for a facility that integrates both thermal wellness AND performance training under one roof.

Created: 2026-02-26 | Updated: 2026-02-26 (v2 — vetted through structured founder review)

Data sources: EXOS facility specs, Othership/Bathhouse/Löyly architecture press, Sauna House locations, Counsilman-Hunsaker pool engineering, GWI Hydrothermal Guide, PoolPak/Dectron HVAC specs, industry standards (StudioGrowth, Merrithew, ElDorado Climbing). Full research briefs available.

Next step: Review with Ramsey Ball at March 5 follow-up meeting.